Arch Linux Installation and Configuration

Platform: VMware virtual machine | **Architecture:** x86-64 | **OS:** Arch Linux | **Desktop:** KDE Plasma | **Primary user:** bradley

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1. Setting Up the Virtual Machine

The installation used a VMware virtual machine with a 50 GB SCSI virtual disk, 2 GB RAM, and 2 virtual processors. The machine boots in legacy BIOS mode rather than UEFI. VMware networking was configured and later verified using the ens33 interface. The VMware settings are shown in Figure 1.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS
lsblk -f
ip link show
```

These commands identify the disk, partitions, mounts, and network interfaces.

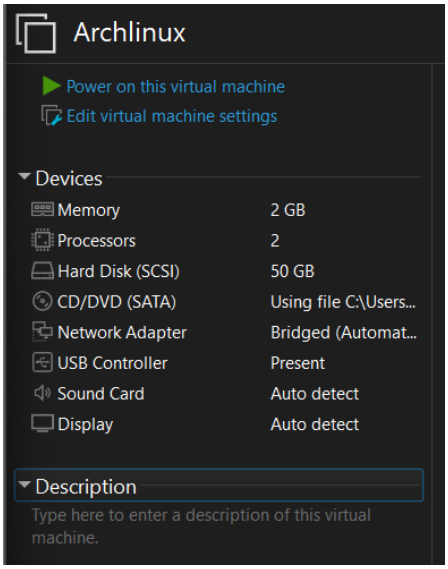


Figure 1. VMware VM configuration: 2 GB memory, 2 processors, 50 GB SCSI disk.

2. Partitioning and Mounting the Disk

The 50 GB virtual disk was /dev/sda. It used an MBR (dos) partition table: /dev/sda1 (~1 GB FAT32, mounted at /boot), /dev/sda2 (~5 GB swap), and /dev/sda3 (~44 GB root filesystem). Unlike the ARM template, this was not an EFI/GPT installation.

```
lsblk -f
mount /dev/sda3 /mnt
mkdir -p /mnt/boot
mount /dev/sda1 /mnt/boot
swapon /dev/sda2
```

The mount commands reflect the intended installation layout. The mount order matters because the bootloader configuration must be generated with /boot mounted.

```
root@archiso ~ # ls /sys/firmware/efi
ls: cannot access '/sys/firmware/efi': No such file or directory
2 root@archiso ~ # sudo fdisk -l
Disk /dev/sda: 50 GiB, 53687091200 bytes, 104857600 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcf7b7810

Device     Boot    Start       End  Sectors  Size Id Type
/dev/sda1           2048   2099199  2097152    1G ef EFI (FAT-12/16/32)
/dev/sda2        2099200  12584960 10485761    5G 82 Linux swap / Solaris
/dev/sda3       12587008 104857599 92270592   44G 83 Linux


Disk /dev/loop0: 1018.84 MiB, 1068331008 bytes, 2086584 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

Figure 2. Partition and disk information recorded during setup.

3. Installing the Base System

The system uses standard Arch Linux for x86-64, not Arch Linux ARM. During troubleshooting I inspected the installed system from the live environment using arch-chroot and checked the kernel files and operating-system identity.

```
arch-chroot /mnt
cat /etc/os-release
ls -lah /mnt/boot
pacman -Q linux
```

The screenshot shows the Arch Linux release details inside chroot.

```
[root@archiso /]# arch-chroot /mnt
bash: arch-chroot: command not found
[root@archiso /]# command -v arch-chroot
[root@archiso /]# cat /etc/os-release
NAME="Arch Linux"
PRETTY_NAME="Arch Linux"
ID=arch
BUILD_ID=rolling
ANSI_COLOR="38;2;23;147;209"
HOME_URL="https://archlinux.org/"
DOCUMENTATION_URL="https://wiki.archlinux.org/"
SUPPORT_URL="https://bbs.archlinux.org/"
BUG_REPORT_URL="https://gitlab.archlinux.org/groups/archlinux/-/issues"
PRIVACY_POLICY_URL="https://terms.archlinux.org/docs/privacy-policy/"
LOGO=archlinux-logo
[root@archiso /]# ls /usr/bin/arch-chroot
ls: cannot access '/usr/bin/arch-chroot': No such file or directory
[root@archiso /]#
```

Figure 3. Checking the installed Arch Linux system in chroot.

4. Configuring the System, Users, and Networking

The account bradley was configured as an administrative user with wheel membership and sudo permissions, and also created codi. Membership in wheel is not automatic: the sudoers rule grants privileges only to users already in that

group. The recorded setup verified bradley's home directory and sudo access; codi's exact group membership should be verified independently.

```
id bradley
id codi
getent passwd bradley codi
sudo whoami
sudo visudo
```

```
%wheel ALL=(ALL:ALL) ALL
```

The network adapter ens33 initially remained DOWN. The setup enabled systemd-networkd and systemd-resolved and configured DHCP with fixed DNS resolvers. A typo (DDHCP instead of DHCP) was corrected.

```
sudo ip link set ens33 up
sudo nano /etc/systemd/network/20-wired.network
```

```
[Match]
Name=ens33

[Network]
DHCP=yes
DNS=1.1.1.1
DNS=1.0.0.1
Domains=~.

[DHCPv4]
UseDNS=no
```

```
sudo systemctl enable --now systemd-networkd
sudo systemctl enable --now systemd-resolved
sudo ln -sf /run/systemd/resolve/stub-resolv.conf /etc/resolv.conf
ping -c 3 1.1.1.1
ping -c 3 archlinux.org
```

Both IP connectivity and DNS resolution were confirmed, including after a reboot.

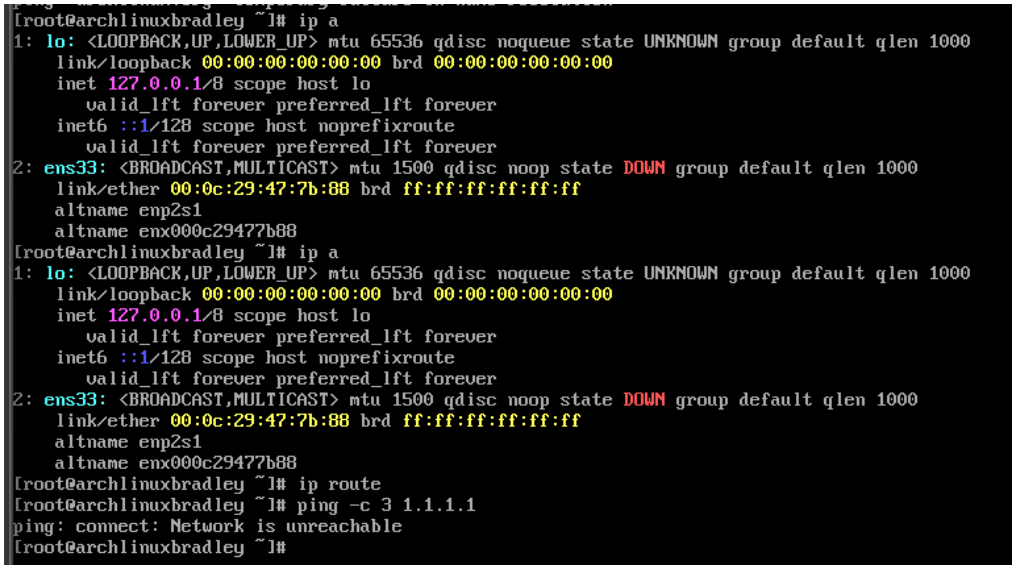


Figure 4. Network interface and connectivity troubleshooting.

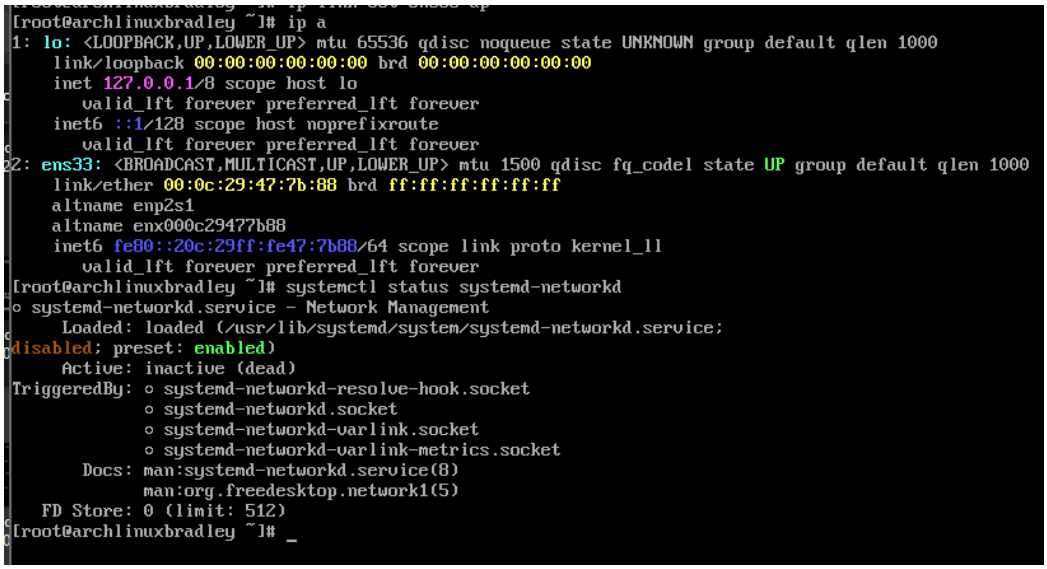


Figure 5. Network configuration and DNS diagnostics.

5. Installing KDE Plasma Desktop

After the base system booted successfully, KDE Plasma was installed rather than LXQt. SDDM provides the graphical login screen; Konsole is the terminal and Dolphin is the file manager.

```
sudo pacman -Syu
sudo pacman -S plasma-meta sddm konsole dolphin
sudo systemctl enable sddm.service
sudo systemctl set-default graphical.target
systemctl get-default
sudo reboot
```

The package installation produced no reported errors. After reboot, the KDE Plasma desktop launched successfully. The default boot target was configured as graphical.target.

6. Configuring and Repairing the Bootloader

This VM uses BIOS boot with GRUB, not the template's UEFI systemd-boot. The initial installation reached a GRUB command prompt instead of the normal boot menu because the boot configuration had been generated without the intended /boot partition mounted. After mounting the partitions and regenerating GRUB, the installed OS booted successfully.

```
mount /dev/sda3 /mnt
mount /dev/sda1 /mnt/boot
arch-chroot /mnt
grub-install --target=i386-pc --boot-directory=/boot --recheck /dev/sda
grub-mkconfig -o /boot/grub/grub.cfg
ls -lah /boot
```

The GRUB configuration then detected the Linux kernel and initramfs. The installation ISO was disconnected for the normal hard-disk boot.

```
root@archiso ~ # arch-chroot /mnt pacman -Q linux
linux 7.2.11.arch1-1
root@archiso ~ # ls -lah /mnt/boot
total 16K
drwxr-xr-x  4 root root 4.0K Nov  1 19:45 .
dr-xr-xr-x 17 root root 4.0K Oct 28 22:40 ..
drwxr-xr-x  2 root root 4.0K Nov  1 19:31 efi-check
drwxr-xr-x  6 root root 4.0K Nov  1 19:45 grub
root@archiso ~ #
```

Figure 6. Boot partition and GRUB directory inspection.

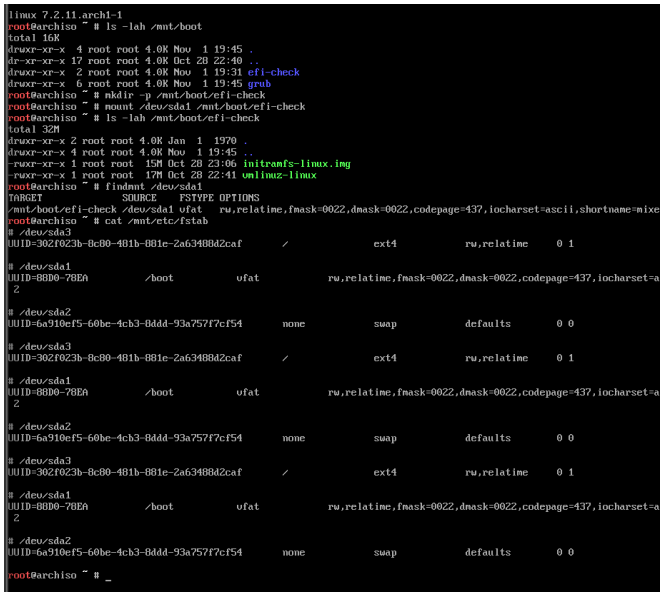


Figure 7. GRUB and mount troubleshooting output.

7. Installing Zsh, SSH, Terminal Colors, and Aliases

The assignment requires an additional shell, not necessarily replacing Bash as the default. Zsh was installed, but continued using Bash as the login shell. Both Zsh and Bash were customized with green username/hostname and blue working-directory prompts.

```
sudo pacman -S zsh
zsh --version
nano ~/.zshrc

autoload -Uz colors && colors
PROMPT='%F{green}%n@%m%f:%F{blue}%~%f %# '
alias ls='ls --color=auto'
alias grep='grep --color=auto'
```

For Bash, The following prompt was added to ~/.bashrc and loaded the changes using source.

```
nano ~/.bashrc
PS1='\[\e[1;32m\]\u@\h\[\e[0m\]:\[\e[1;34m\]\w\[\e[0m\]\$ '
alias ls='ls --color=auto'
alias grep='grep --color=auto'
source ~/.bashrc
```

The OpenSSH server was installed and enabled, then tested connecting locally.

```
sudo pacman -S openssh
sudo systemctl enable --now sshd
systemctl status sshd
ssh bradley@localhost
exit
```

Several Bash aliases were added. Initial attempts to use ll and myip failed because the configuration had not yet been sourced. After running source ~/.bashrc, the shortcuts worked.

```
alias ll='ls -lah --color=auto'
alias la='ls -A --color=auto'
alias ..='cd ..'
alias ...='cd ../..'
alias update='sudo pacman -Syu'
alias pkgsearch='pacman -Qs'
alias myip='ip -br addr'
alias cls='clear'
alias sysinfo='uname -a'
```

```
source ~/.bashrc
ll
myip
sysinfo
alias
```

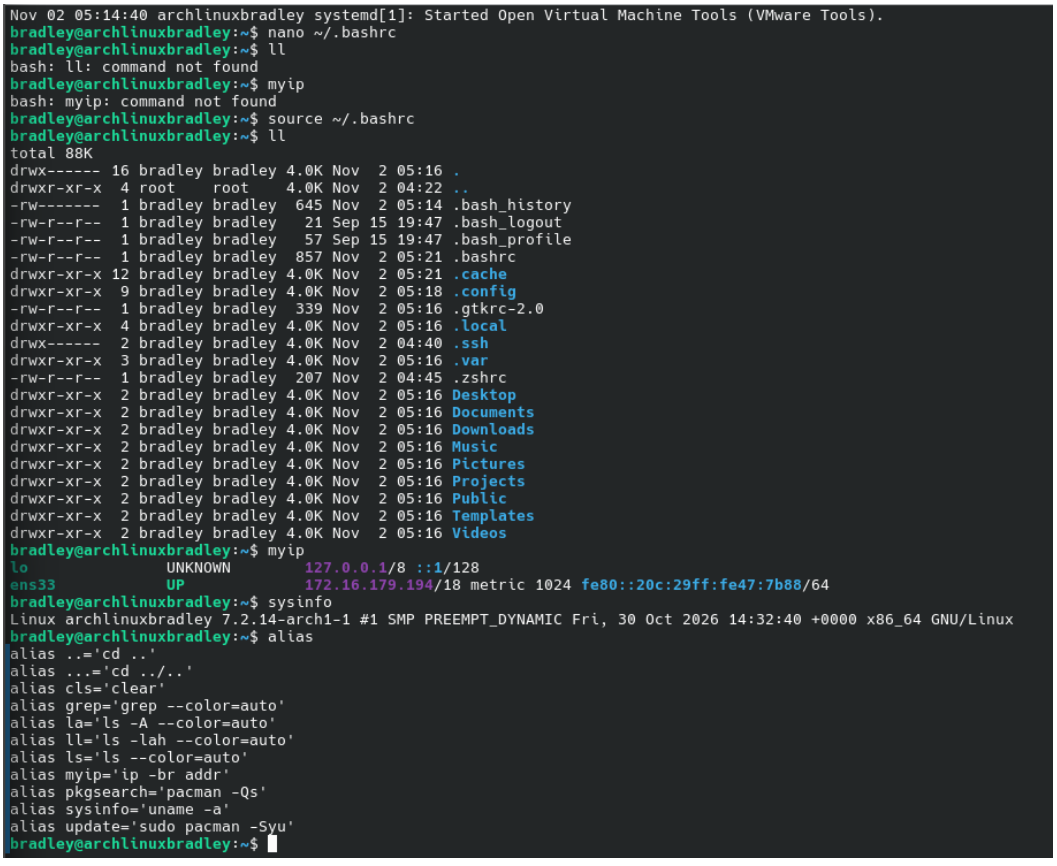


Figure 8. Successful Bash color prompt and aliases; ens33 has 172.16.179.194.

8. VMware Tools and Final Testing

To improve VMware guest integration, I installed open-vm-tools and enabled vmtoolsd. The service was confirmed active (running). The screenshot of the terminal shows the VMware Tools startup message, colored prompt, successful aliases, and Arch Linux kernel details.

```
sudo pacman -S open-vm-tools
sudo systemctl enable --now vmtoolsd.service
systemctl status vmtoolsd.service
systemctl get-default
ps -p $$ -o comm=
echo $SHELL
ip -br addr
```

At the time of the screenshot, ens33 was UP with address 172.16.179.194/18. Bash remained the active/default shell while Zsh was installed separately. The GUI login and desktop were confirmed interactively, although no screenshot of the running KDE desktop was supplied.

9. Problems Encountered and Resolutions

Problem 1 — GRUB opened to a command prompt instead of booting Arch Linux. The initial GRUB configuration was generated while the intended /boot partition was not mounted. As a result, the boot menu did not locate the installed kernel and initramfs. The root and boot partitions were remounted in the correct order, the installed system was entered with arch-chroot, and GRUB was reinstalled and regenerated. The configuration then detected the kernel and initramfs, and Arch Linux booted from the virtual disk.

```
mount /dev/sda3 /mnt
mount /dev/sda1 /mnt/boot
arch-chroot /mnt
grub-install --target=i386-pc --boot-directory=/boot --recheck /dev/sda
grub-mkconfig -o /boot/grub/grub.cfg
```

Problem 2 — Package updates failed in the live installation environment. The system was still running from the Arch installation ISO (airootfs), which had insufficient writable space for the attempted package operation. The resolution was to boot the installed Arch system rather than continue updating the live ISO. Package management then proceeded in the installed operating system.

Problem 3 — The VMware network interface was down. The ens33 interface was initially DOWN, preventing network connectivity. It was brought up manually, and systemd-networkd was enabled with a DHCP configuration so connectivity could persist after reboot.

```
ip link set ens33 up
systemctl enable --now systemd-networkd
ip -br addr
```

Problem 4 — DHCP failed because of a configuration typo. The network file mistakenly contained DDHCP instead of DHCP. The directive was corrected to DHCP=yes, and network services were used to obtain an address. The final configuration specified Cloudflare DNS servers and disabled DHCP-provided DNS; successful pings to both an IP address and archlinux.org confirmed routing and name resolution.

```
nano /etc/systemd/network/20-wired.network
# Correct directive: DHCP=yes
ping -c 3 1.1.1.1
ping -c 3 archlinux.org
```

Problem 5 — DNS resolution needed persistent configuration. systemd-resolved was enabled, and /etc/resolv.conf was linked to its stub resolver. This provided a persistent resolver configuration after restarting the virtual machine.

```
systemctl enable --now systemd-resolved
ln -sf /run/systemd/resolve/stub-resolv.conf /etc/resolv.conf
```

Problem 6 — Zsh was installed, but Bash remained the login shell. The assignment required an additional shell, not a replacement default. Zsh installation was verified, while Bash remained the default. To ensure the active terminal displayed colors, the prompt was also configured in ~/.bashrc rather than relying only on ~/.zshrc.

```
zsh --version
echo $SHELL
ps -p $$ -o comm=
nano ~/.bashrc
```

Problem 7 — Custom aliases initially returned "command not found." The aliases had been written to ~/.bashrc but had not yet been loaded in the active Bash session. Running source ~/.bashrc immediately activated them. The ll, myip, and sysinfo commands then produced the expected directory listing, interface addresses, and kernel information, as shown in the terminal screenshot.

```
source ~/.bashrc
ll
myip
sysinfo
alias
```

Conclusion

The Arch Linux installation successfully booted in VMware after GRUB repair. Persistent networking, user accounts and sudo, Zsh, OpenSSH, colored Bash prompts and aliases, KDE Plasma graphical startup, and VMware Guest Tools were configured. The terminal screenshots confirm working aliases, network connectivity, and the configured terminal prompt.

References

ArchWiki — Installation Guide

ArchWiki — GRUB

ArchWiki — KDE Plasma

ArchWiki — SDDM

ArchWiki — OpenSSH

ArchWiki — systemd-networkd

ArchWiki — Zsh

ArchWiki — Bash